

Overview

The purpose of this exercise is to review and analyze recommender applications using a recommender system taxonomy. We chose three platforms (Spotify, Netflix, Zalando) and for each platform we identified two different recommender techniques and classified them along the taxonomy dimensions.

1 Spotify

1.1 Technique 1: “Popular with listeners of ...” (artist/track-based suggestions)

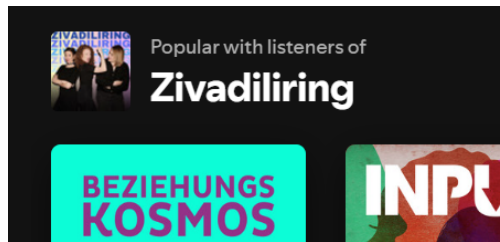


Figure 1: Spotify: “Popular with listeners of ...”

Domain	Digital music and podcast streaming platform where artists upload music/podcasts and users consume content via subscription (or a free version with ads). Builds a large user community around music.
Purpose of recommendation	Help users discover music and podcasts they likely enjoy; build a community around listeners of similar items; reduce search effort and decision fatigue; encourage exploration of new artists within genres of interest.
Recommendation context	Appears while browsing/listening and reacts directly to the current listening choice (context-dependent). The suggestion is linked to the current artist/track and to what similar listeners typically play next.
Whose opinion	Primarily “people like you” (users with similar listening behaviour). This matches a collaborative-filtering intuition; editorial and sponsored influence may also exist.
Personalization level	Tailored to the individual user. The recommendation depends on the specific artist/track and similar users’ behaviour. Ephemeral component: the queue/suggestions can change immediately when switching to a different artist or track.
Privacy and trustworthiness	Relatively non-sensitive personal data (music taste, listening behaviour). Commercial influence (e.g., sponsored recommendations) may exist but potential harm is limited; manipulation would typically not expose highly sensitive information.
Interfaces	Input: explicit signals such as saving/liking items and (during sign-up) selecting favourite artists/genres; implicit signals such as skips and follows. Output: recommended list (ranked suggestions).
Recommendation algorithm	Collaborative filtering (items popular among similar listeners). It may also include content-based signals such as similarity in genre/artist.

1.2 Technique 2: “Trending songs” (popularity / location-aware lists)

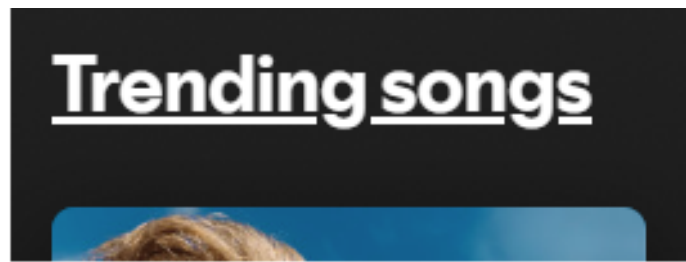


Figure 2: Spotify: “Trending songs”

Domain	Digital music and podcast streaming platform; music discovery and consumption (including playlists).
Purpose of recommendation	Keep users engaged by suggesting relevant and currently popular content (trending music), which supports advertising revenue (free tier) and increases listening time. Helps users quickly discover widely consumed music.
Recommendation context	Appears as trend lists; seems context-aware in the sense that it can recommend items based on user location and local/global trends. Trends change frequently.
Whose opinion	Aggregated behaviour of many users (popularity). This can be interpreted as “people like you” at a broad aggregate level; editorial and sponsored influence may exist.
Personalization level	Generally low personalization (global trends) and low-to-medium personalization when country-based. For logged-in users, trending content may be combined with location and individual behaviour. Strongly time-ephemeral because trending songs change frequently.
Privacy and trustworthiness	Relatively non-sensitive data (location and listening behaviour) used in aggregate. Potential commercial influence exists, but the harm is limited.
Interfaces	Input: implicit behaviour (streams, skips/ignores) aggregated across users (often by location); explicit through listening to artists or saving music/podcasts. Output: ranked list of trending items.
Recommendation algorithm	Popularity-based ranking to identify most consumed songs; potentially refined with collaborative-filtering signals and context (current location).

2 Netflix

2.1 Technique 1: “Top 10 TV Shows / Movies in Switzerland” (regional popularity list)

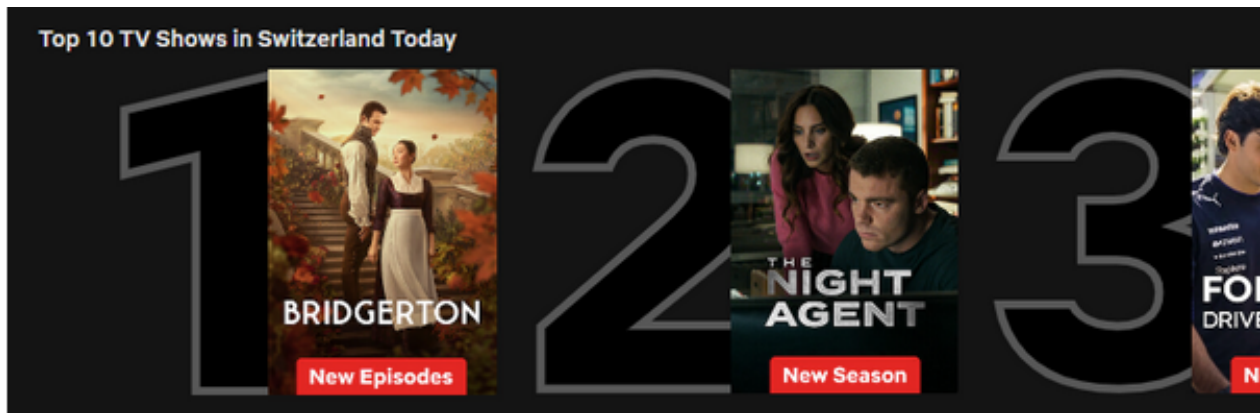


Figure 3: Netflix: “Top 10 TV Shows in Switzerland Today”

Domain	Digital video streaming platform providing films, series and documentaries via subscription; connects content producers and viewers.
Purpose of recommendation	Help users quickly find what to watch by showing popular content within a regional audience; reduce search effort and decision fatigue; encourage engagement and shared viewing trends.
Recommendation context	Strongly context-based on the user’s current location (e.g., Switzerland). By showing what is popular locally, Netflix encourages trend-following and social discussion, which can increase overall screen time.
Whose opinion	“People like you” at a regional level: aggregated viewing behaviour of users in the same country/region. Ranking is driven by viewing numbers.
Personalization level	Low personalization: all users in Switzerland can see the same Top 10 list regardless of individual taste. Time-ephemeral: the list can change daily.
Privacy and trustworthiness	Low risk because it is based on aggregated behaviour of a whole country. The list may influence viewing, but manipulation risk is limited because users can skip content. Platform-owned productions may appear prominently and this can be visible to users.
Interfaces	Input: explicit user interaction, namely the items that the user chooses to watch. Additionally, implicit feedback is generated through user behaviour. For example, starting a highly ranked title and stopping it midway (e.g., due to lack of interest) may influence future recommendations and ranking adjustments. Output: explicitly displayed ranked list.
Recommendation algorithm	Context-aware (location-based) popularity ranking: viewing counts within Switzerland are analysed and combined into a Top 10 list.

2.2 Technique 2: “Because you watched ...” (personalized similar-item suggestions)

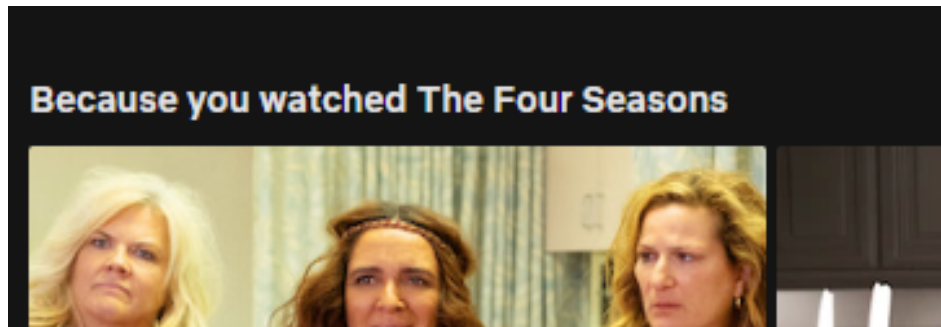


Figure 4: Netflix: “Because you watched ...”

Domain	Digital video streaming platform (movies, series, documentaries) consumed via subscription.
Purpose of recommendation	Offer highly relevant recommendations that match the user’s taste; reduce search effort by recommending based on the user’s watching behaviour; keep the user engaged and increase watching time; improve satisfaction by matching preferred genres/themes.
Recommendation context	Recommendations appear directly after watching an item and/or on the personalized homepage. The goal is to keep the user engaged and increase viewing time.
Whose opinion	Primarily “people like you”: uses behaviour of users with similar tastes and how they reacted to the same content. Experts play a minimal role for this technique.
Personalization level	High personalization: each user has a distinct homepage. Strongly ephemeral: suggestions can change soon after watching, searching, or skipping items; “You might also like ...” appears immediately after finishing a title.
Privacy and trustworthiness	Moderate risk: based on entertainment preferences (watch history, genre interests). Manipulation risk is limited since users can stop watching. Platform-owned productions may appear higher, which can be noticeable to users.
Interfaces	Input: explicit from watched items and search behaviour/items; additional implicit feedback from skipping suggestions or choosing lower-ranked items. Output: ranked list of items the user is likely to enjoy.
Recommendation algorithm	Hybrid approach: collaborative filtering (similar viewing behaviour) plus content-based techniques (genre/themes such as thrillers, horror). May also incorporate contextual factors (e.g., region lists), but the core is personalized similarity.

3 Zalando

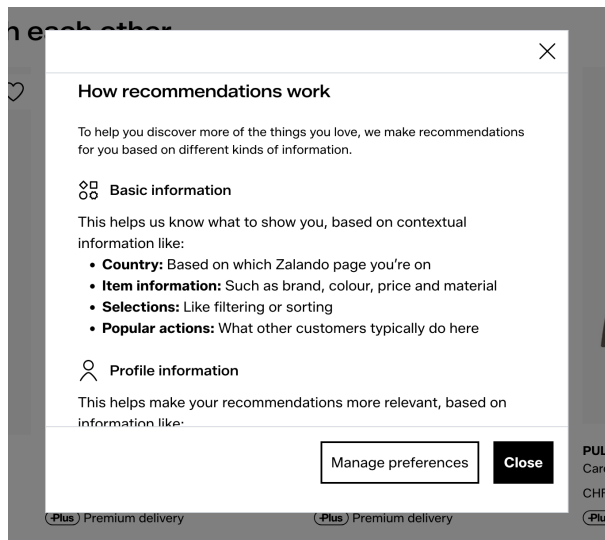


Figure 5: *

Zalando: “How recommendations work” (basic information)

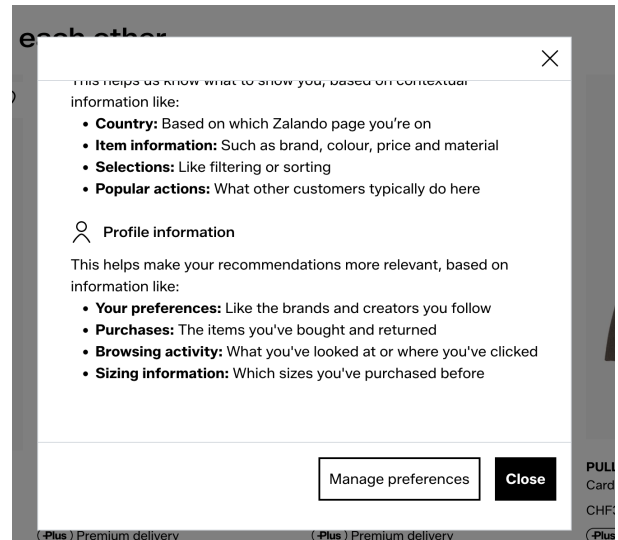


Figure 6: *

Zalando: “How recommendations work” (profile information)

Figure 7: Zalando transparency: explanation of contextual and profile-based recommendation signals

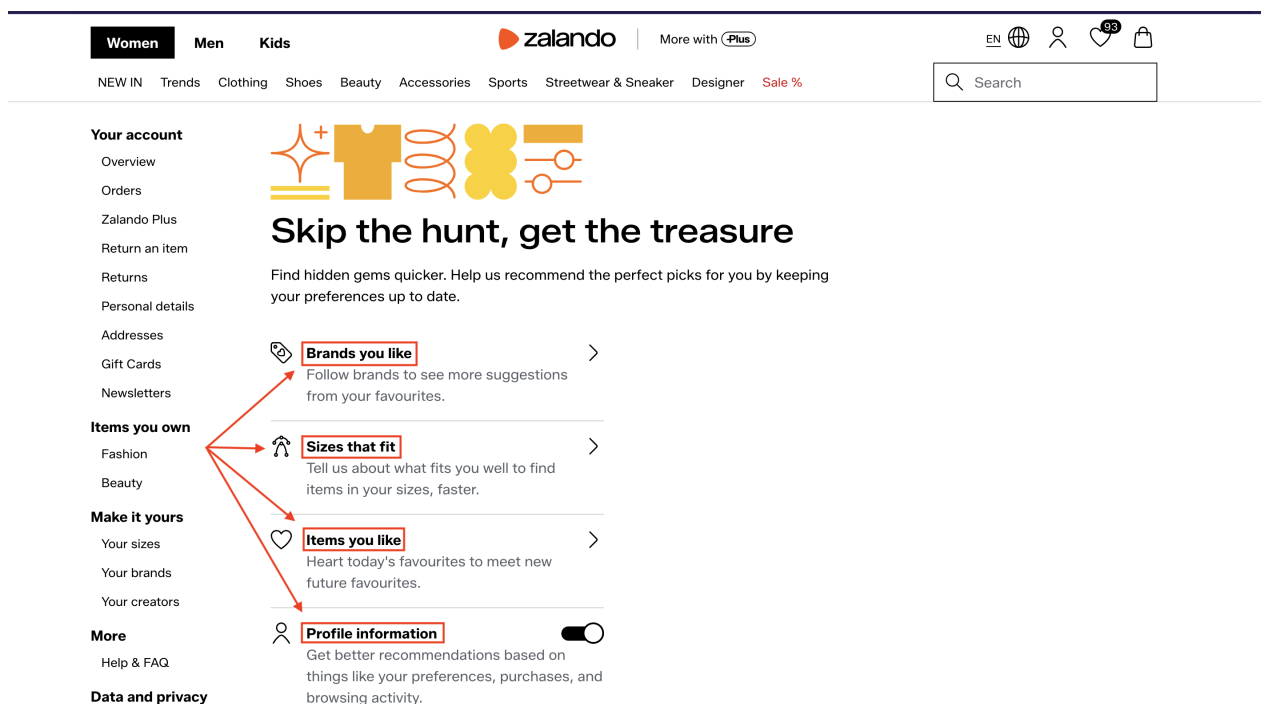


Figure 8: Zalando: profile settings (brands you like, sizes that fit, items you like, profile information)

3.1 Technique 1: “Recommended for you” (homepage personalization)

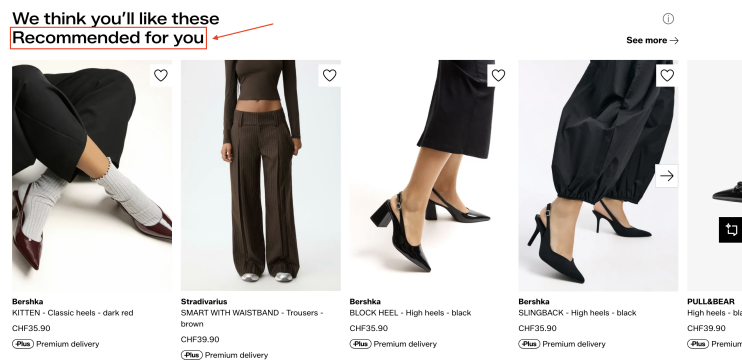


Figure 9: Zalando: “Recommended for you”

Domain	Products (fashion items: clothing, shoes, accessories).
Purpose of recommendation	Encourage product discovery; increase engagement; increase sales; reduce search effort and decision fatigue; personalize the shopping experience.
Recommendation context	Displayed on the homepage for logged-in users under “Recommended for you”. Based on the platform’s explanation of how recommendations work, contextual signals include: country (which Zalando page you are on), item information (brand, colour, price, material), selections (filtering/sorting), and “popular actions” (what other customers typically do). Profile-related signals can include preferences, purchases/returns, browsing activity, and sizing information.
Whose opinion	Primarily “people like you”: the use of “popular actions” and similarity in purchasing/browsing behaviour indicates collaborative signals. The system is not primarily expert-driven; commercial promotion may influence visibility.
Personalization level	High personalization. Combination of persistent personalization (purchase history, browsing activity, size preferences) and ephemeral personalization (recent clicks and filtering/sorting). Country-level adaptation can also be part of the context.
Privacy and trustworthiness	Medium to high risk: uses shopping behaviour signals such as purchases (including returns), browsing activity, sizing information, preferences, and country. This can reveal lifestyle preferences or socioeconomic patterns. Transparency is improved by explaining what signals are used; commercial bias is possible.
Interfaces	Input: implicit (clicks, views, purchases, cart activity, browsing); explicit (saved/liked items, followed brands, size preferences). Output: personalized ranked product lists and suggested categories on the homepage.
Recommendation algorithm	Hybrid recommender system: collaborative filtering (behaviour of similar customers) + content-based filtering (product attributes such as brand/colour/price/material) + context-aware elements (country and user selections like filters).

3.2 Technique 2: “Better together – Often viewed with each other” (product-page association)

Better together

Often viewed with each other

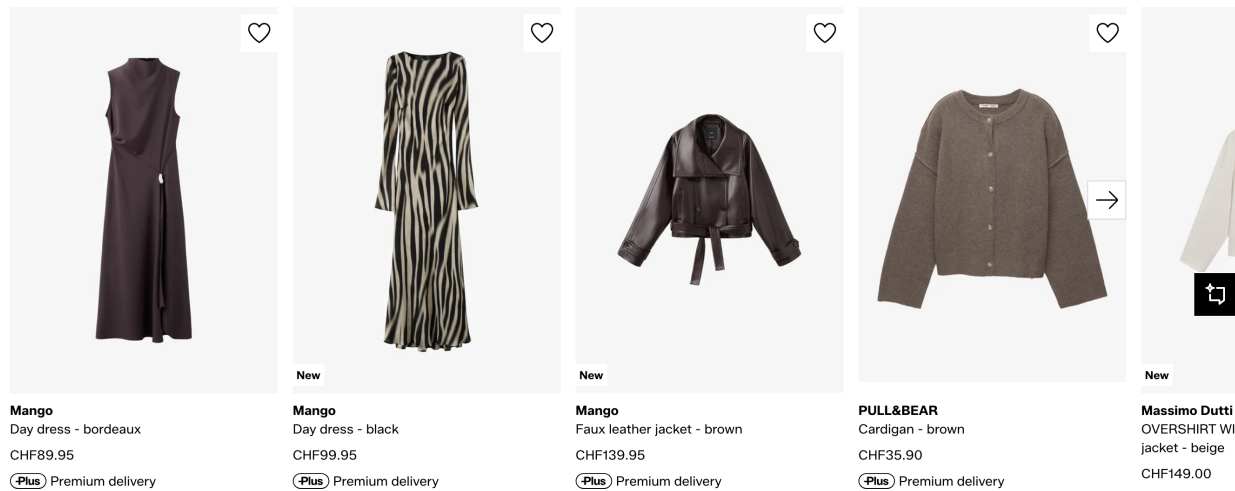


Figure 10: Zalando: “Better together – Often viewed with each other”

Domain	Products (fashion items: clothing, shoes, accessories).
Purpose of recommendation	Increase cross-selling; encourage outfit completion; increase basket value by suggesting complementary items.
Recommendation context	Displayed on product pages under “Better together – Often viewed with each other”. Triggered by viewing a specific product; the recommendation is strongly item-based and depends on the currently viewed item.
Whose opinion	Primarily “people like you” in aggregate: “often viewed with each other” indicates co-view/co-purchase patterns from many users rather than expert curation.
Personalization level	Mostly ephemeral and item-based (dependent on the current product). Less dependent on long-term user profile than homepage personalization.
Privacy and trustworthiness	Medium risk: relies mainly on aggregated behavioural patterns (co-views/co-purchases). Involves less personal profiling than homepage personalization; commercial bias may still exist.
Interfaces	Input: implicit behaviour signals (views, clicks, purchases) aggregated across users. Output: complementary product suggestions/outfit combinations.
Recommendation algorithm	Likely item-item collaborative filtering and/or association-rule style co-occurrence statistics (“viewed together” patterns).